Finding Parameters That Make Piecewise Functions Continuous
Answer Key

AP Calculus AB and BC   Limits and Continuity

Quick Answers

1. -1
2. 6
3. 2
4. a = 3, b = -2
5. $\frac{1}{6}$
6. 2
7. a = -1, b = -4
8. —

Curriculum

Level: AP Calculus AB and BC   Limits and Continuity
LIM-1, LIM-2 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

2. *If they got 0: cancelled the wrong factor and evaluated x - 3 at x = 3 instead of x + 3*
5. *If they got 0: substituted x = 2, saw the numerator go to 0, and reported the limit as 0 without resolving the 0/0 form*
6. *If they got 4: read the limit straight off the exponent and ignored the 2 in the denominator*

1. $f(x) = 3x + a$ for $x < 2$, $x^2 + 1$ for $x \geq 2$. Find a.

Step 1. Both pieces are polynomials, so each is continuous on its own side. The only place continuity can fail is the junction $x = 2$, and there the test reduces to: left-hand limit = right-hand limit = $f(2)$.
Step 2. $\lim_{x\to 2^-} f(x) = 3(2) + a = 6 + a$, $\lim_{x\to 2^+} f(x) = 2^2 + 1 = 5 = f(2)$.
Step 3. Set them equal: $6 + a = 5$, so $\boxed{a = -1}$
Check: with $a = -1$ the left piece gives $3(2) - 1 = 5$, matching the right piece.

2. $f(x) = \dfrac{x^2 - 9}{x - 3}$ for $x \neq 3$, k at $x = 3$. Find k.

Step 1. Direct substitution gives $\frac{0}{0}$, an indeterminate form, so the limit is not read off by plugging in — factor and cancel the common factor first.
Step 2. $\dfrac{x^2 - 9}{x - 3} = \dfrac{(x - 3)(x + 3)}{x - 3} = x + 3$ for every $x \neq 3$, so $\lim_{x\to 3} f(x) = 3 + 3 = 6$.
Step 3. Continuity at $x = 3$ requires $f(3) = \lim_{x\to 3} f(x)$, so $\boxed{k = 6}$
Common wrong answer: 0 — the wrong factor was cancelled and $x - 3$ was evaluated at $x = 3$. Cancel $(x - 3)$, then substitute into what is left.

3. $T(t) = 4t + c$ **on** $0 \le t < 5$**,** $32 - 2t$ **for** $t \ge 5$**. Find** c **in degrees Celsius.**

Step 1. The model must not jump at the moment the two rules change over, so the two one-sided limits at $t = 5$ minutes have to agree.

Step 2. From the left: $4(5) + c = 20 + c$ degrees Celsius. From the right: $32 - 2(5) = 22$ degrees Celsius, which is also $T(5)$.

Step 3. $20 + c = 22$, so $\boxed{c = 2}$

Interpretation: the oven is already at 2°C above ambient when the timer starts; without that offset the model would drop instantaneously by 2°C at the five-minute mark.

4. $f(x) = x^2$ **on** $x \le 1$**,** $ax + b$ **on** $1 < x < 4$**,** 10 **on** $x \ge 4$**. Find** a **and** b**.**

Step 1. There are two junctions, $x = 1$ and $x = 4$, and each one gives one equation — two equations for the two unknowns, so the system is determined.

Step 2. At $x = 1$: the left piece gives $1^2 = 1$ and the middle piece approaches $a(1) + b$, so $a + b = 1$.

Step 3. At $x = 4$: the middle piece approaches $a(4) + b$ and the right piece is 10, so $4a + b = 10$.

Step 4. Subtract the first equation from the second: $3a = 9$, so $a = 3$, and then $b = 1 - 3 = -2$. $\boxed{a = 3,\ b = -2}$

Check: $3(1) - 2 = 1$ matches x^2 at $x = 1$, and $3(4) - 2 = 10$ matches the constant piece.

5. $f(x) = \dfrac{\sqrt{x+7} - 3}{x - 2}$ **for** $x \ne 2$**,** k **at** $x = 2$**. Find** k **exactly.**

Step 1. Substitution gives $\frac{0}{0}$ again, but nothing factors — with a radical in the numerator the tool is the conjugate, which converts the difference of square roots into a difference of squares.

Step 2. Multiply numerator and denominator by $\sqrt{x+7} + 3$:

$$\frac{(\sqrt{x+7} - 3)(\sqrt{x+7} + 3)}{(x-2)(\sqrt{x+7} + 3)} = \frac{(x+7) - 9}{(x-2)(\sqrt{x+7} + 3)} = \frac{x - 2}{(x-2)(\sqrt{x+7} + 3)}.$$

Step 3. Cancel $x - 2$ and substitute: the limit is $\dfrac{1}{\sqrt{9} + 3} = \dfrac{1}{6}$, so $\boxed{k = \frac{1}{6}}$

Common wrong answer: 0 — the numerator tends to 0, but so does the denominator, and a $\frac{0}{0}$ form has to be resolved before any value can be read off.

6. $f(x) = \dfrac{e^{4x} - 1}{2x}$ **for** $x \ne 0$**,** k **at** $x = 0$**. Find** k**.**

Step 1. Substitution gives $\frac{0}{0}$. The numerator is the difference quotient shape for e^{4x} at $x = 0$, so either the known limit $\lim_{u \to 0} \frac{e^u - 1}{u} = 1$ or L'Hôpital's rule settles it.

Step 2. Write $u = 4x$: $\dfrac{e^{4x} - 1}{2x} = \dfrac{e^u - 1}{u} \cdot \dfrac{4x}{2x} = \dfrac{e^u - 1}{u} \cdot 2 \to 1 \cdot 2.$

Step 3. The limit is 2, and continuity forces $f(0)$ to equal it, so $\boxed{k = 2}$

Common wrong answer: 4 — the exponent was read as the answer and the 2 in the denominator was ignored.

7. $f(x) = 2x + a$ **for** $x < 3$, $x^2 + b$ **on** $3 \le x < 5$, $4x + 1$ **for** $x \ge 5$. **Find** a **and** b.
Step 1. Two junctions again, but they are not symmetric: the junction at $x = 5$ involves only b, so solve that one first and carry the value into the other equation.
Step 2. At $x = 5$: the middle piece approaches $5^2 + b = 25 + b$ and the right piece gives $4(5) + 1 = 21$. So $25 + b = 21$ and $b = -4$.
Step 3. At $x = 3$: the left piece approaches $2(3) + a = 6 + a$ and the middle piece gives $3^2 + b = 9 + b = 9 - 4 = 5$. So $6 + a = 5$.
Step 4. Therefore $\boxed{a = -1,\ b = -4}$
Check: the middle piece runs from $f(3) = 9 - 4 = 5$ up to $5^2 - 4 = 21$, meeting both neighbours with no jump.

8. Why no k makes $f(x) = \dfrac{1}{x-3}$ (with $f(3) = k$) continuous at $x = 3$. (Manual review — open reasoning.)
Model answer. Continuity at $x = 3$ requires $\lim_{x\to 3} f(x)$ to exist *and* to equal $f(3)$. Examine the two one-sided limits.
Step 1. For x slightly less than 3, $x - 3$ is a small negative number, so $\dfrac{1}{x-3} \to -\infty$: $\lim\limits_{x\to 3^-} f(x) = -\infty$.
Step 2. For x slightly greater than 3, $x - 3$ is a small positive number, so $\lim\limits_{x\to 3^+} f(x) = +\infty$.
Step 3. Neither one-sided limit is a real number, so $\lim_{x\to 3} f(x)$ does not exist. A value of k can only supply $f(3)$; it cannot create a limit that is not there. Hence $\boxed{\text{no such } k \text{ exists}}$
Step 4. Because at least one one-sided limit is infinite, $x = 3$ is an *infinite* (vertical-asymptote) discontinuity, which is non-removable. The student's $k = 5$ fixes nothing: f still fails the limit condition.
Grading note: full credit needs both one-sided limits stated (not just "the function blows up"), the conclusion that the two-sided limit does not exist, and the word *infinite* or *non-removable* for the classification.